

Running Gradient Descent: The Algorithm in Action

This section visualizes the step-by-step process of gradient descent finding the optimal parameters for a linear regression model.

The Iterative Process Visualized

1. **Initialization:** We start with an initial guess for the parameters w and b . This corresponds to a specific starting point on the cost function's contour plot and a poorly fitting line on the data plot.
 2. **Taking Steps:**
 - With each iteration (or step) of gradient descent, the algorithm calculates the direction of steepest descent and updates w and b .
 - On the contour plot, you can see the point representing (w, b) move closer to the center (the minimum).
 - Simultaneously, on the data plot, the straight line $f(x)$ adjusts its slope and intercept, providing a slightly better fit to the data with each step.
 3. **Convergence:**
 - This process is repeated, with the cost $J(w, b)$ decreasing at each step.
 - Eventually, the algorithm **converges** at the **global minimum**—the center of the contour plot.
 - The final w and b values at this minimum define the model that best fits the training data. This final model can then be used to make predictions on new data.
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Batch Gradient Descent

- The specific version of the algorithm we have been using is called **Batch Gradient Descent**.
 - **Definition:** The term "batch" means that on **every single step** of gradient descent, we look at the **entire training set** (all m examples) to calculate the derivatives. This is reflected in the summation term, $\sum_{i=1}^m$, which runs over all the examples.
 - *Note:* Other versions of gradient descent exist that look at smaller subsets of data, but for now, we are using the "batch" approach.
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